

UNIVERSE TUTORIAL 07/07/2026 (HR & JPC)

THEORETICAL BACKGROUND

Our current understanding of the Universe on large scales is based on:

- General Relativity, the geometric theory of gravitation that relates variations in the geometry of space-time to variations in its matter-energy content.
- The Cosmological Principle, according to which the contents of the Universe are homogeneous and isotropic (the same everywhere and in every direction at a given moment).

The resulting model of *Universe* can represent reality only on scales where the real Universe is approximately homogeneous and isotropic. When Einstein published his first static universe model in 1917, the cosmological principle was merely a simplifying assumption, but it has gradually been confirmed by observations throughout the twentieth century. Furthermore, the cosmic microwave background radiation shows that, 380,000 years after the Big Bang, the largest inhomogeneities were only about 10^{-5} (1/100,000).

It is from these tiny inhomogeneities, together with the cooling caused by the expansion of space, that matter was able to form massive objects (galaxies, stars, planets, etc., including black holes). Among these approximately spherical and centrosymmetric objects, the calculation of the orbits of bodies (asteroids, spacecraft, etc.) whose masses are negligible compared with that of the central spherical object can be performed using the *TRAJECTORIES* module of Cosmogravity. For spacecraft, Cosmogravity also provides a flight-control mode.

The Universe has progressively become structured and continues to do so. Simulating these structures with a very large number of cells requires computational power that is currently beyond the capabilities of our PCs and smartphones.

However, as indicated below, many astronomical bodies (planets, stars, and black holes) are highly centrosymmetric. The *TRAJECTORIES* section of Cosmogravity therefore allows the calculation and display of the trajectory of an object whose mass is negligible compared with that of a single massive centrosymmetric object. The moving object may be either natural or artificial (such as a spacecraft, for which a piloting option is available).

Returning to *UNIVERSE*.

The reduced scale factor of the Universe, $a(t)$, is equal to 1 at the present time $t = t_0$. Within the framework of General Relativity and the Cosmological Principle, the curve $a(t)$ describing the reduced scale factor of space is completely determined by the present values of three of the four density parameters Ω_i (where i may represent radiation, matter, the cosmological constant Λ , or curvature k), together with the present value H_0 of the space expansion rate H .

Example Procedure

1. By opening the **Universe > Simulation > Cosmological Constant tab**, the simulation of the standard model (Planck Collaboration 2015 parameter values) is launched, displaying its corresponding $a(t)$ curve.

2. You can modify the Inputs and simulate different universes by changing the values of the density parameters and the expansion rate.

Note: the radiation density parameter Ω_r is calculated from the CMB temperature entered in the input fields. Click "Plot" to launch the new simulation.

3. You can also use the interactive diagram to modify Ω_{m0} and $\Omega_{\Lambda0}$, but first read the *INTERACTIVE DIAGRAM* section below.

4. In every case, you can save both the input parameters and the generated graph by clicking Save.

5. The Flat Universe option forces $\Omega_k = 0$ by adjusting $\Omega_{\Lambda0}$.

Cosmogavity 2026 allows computation of both the standard Λ CDM model and a Dark Energy (DE) model, in which a physical fluid replaces the geometrical cosmological constant Λ , although Λ itself appears as a particular case of Dark Energy.

As of early 2026, the analysis of large-scale observational data is underway to determine which of these descriptions best represents our Universe.

INTERACTIVE DIAGRAM (A GOOD PLACE TO START?)

One of the distinctive features of Cosmogravity UNIVERSE is its interactive diagram. It may be the best way to begin using the UNIVERSE module. However, this convenience requires a few precautions.

By selecting UNIVERSE > SIMULATION, you obtain a page displaying an interactive diagram on the left, with axes Ωm_0 and $\Omega \Lambda_0$.

This page initially displays the density parameters of the standard Λ CDM (Lambda Cold Dark Matter) model, a universe whose contents consist of radiation, non-relativistic matter, the cosmological constant, and spatial curvature.

Near this standard model, the density parameters corresponding to radiation and curvature are very small. Consequently, the two parameters Ωm_0 and $\Omega \Lambda_0$ determine the type of Universe, except in the immediate vicinity of the boundary curves shown on the diagram.

Clicking anywhere in the interactive diagram launches a simulation using the selected values of Ωm_0 and $\Omega \Lambda_0$. However, the code also uses the values of the other two density parameters, Ωr_0 and Ωk_0 , which are taken from (or derived from) the values displayed in the editable input fields.

You may also enter numerical values directly for these two density parameters.

For Ωr_0 , this is done indirectly through the temperature T_0 , since the cosmic microwave background radiation exhibits a black-body spectrum that is extraordinarily close to Planck's law.

As established in the Theory presentation, the sum of the four density parameters is always equal to 1, making it possible to determine all four parameters while directly observing only three of them.

The interactive graph therefore allows you to generate and visualize the graph of the function $a(t)$, the scale factor of the Universe as a function of time t .

The boundary curves shown on this interactive graph are fixed and are strictly exact only for a universe containing matter and the cosmological constant.

The cosmological constant Λ is mathematically equivalent, in Einstein's field equations of General Relativity, to a particular physical fluid—a form of dark energy—whose equation of state relating pressure and mass density is $p = w\rho c^2$ with $w = -1$. Dark Energy, however, may have a value of w that differs from -1 and may even vary with time. Consequently, Λ appears as a special case of Dark Energy.

The boundary curves on the graph are labeled according to the corresponding types of $a(t)$ evolution; these labels are generally self-explanatory. However, it is useful to clarify the meaning of ACCELERATING and DECELERATING, which appear on either side of the corresponding boundary.

- Accelerating:
 d^2a/dt^2 , the second derivative of $a(t)$, is positive at the present time t_0 .
- Decelerating:
 d^2a/dt^2 , the second derivative of $a(t)$, is negative at the present time t_0 (see Hobson, p. 394¹).

COSMOLOGICAL CALCULATOR: The Observer's Toolbox

Clicking Cosmological Calculator opens a new window. The input values from the main window are automatically recalled.

A wide range of observational calculations is available:

- Density calculations
- Apparent diameter
- Inverse calculations
- Cosmological horizons
- Cosmological parameters
- Geometry
- Photometry
- Graph generator (linear and logarithmic scales)
- Universal constants c , G , k , and h , whose values can be modified in Cosmogravity (other universes?)
- ...

¹ M. P. Hobson, G. Efstathiou, and A. N. Lasenby, *General Relativity: An Introduction for Physicists*, first published in 2006, 7th edition (2014), p. 394.